

42 THE FOUNDING OF BAGHDAD

ibn Bukhtishu, Jurjis's grandson, to head the new hospital in Baghdad, the first in the Islamic world.

Al-Mansur established a library in Baghdad to house his collection of manuscripts. This venture was made easier by the Arab adoption of paper as a medium for books, and the establishment in Baghdad in 795 of a paper mill. However, since Arabic speakers had no access to this learning, the library did little to advance an indigenous Arab scientific tradition.

House of Wisdom

To remedy this, Harun al-Rashid (caliph from 786 to 809) and al-Mamun (reigned 813–833) established the *Bayt al Hikma* (House of Wisdom), which not only housed the growing library, but also acted as an academy for scholars and a center for the translation of key scientific works into Arabic. Among its leading scholars were Hunayn ibn Ishaq (808–873), a Nestorian Christian from al-Hira in Iraq, who translated more than 100 mostly medical and philosophical works; and Thabit ibn Qurra, a member of a pagan sect known as the Sabaeans, who translated *Elements*, Euclid's great work on geometry, and the *Almagest*, Ptolemy's key work on astronomy.

Translation became a highly prestigious endeavor. One Arab patron paid an extravagant 2,000 dinars a month to ensure

his association with a translation of a work by the Greek physician Galen (a dinar, made of pure gold, weighed the same as 72 grains of barley). Within around 150 years, almost all of the key Greek texts that had been discovered had been rendered into Arabic. Many of them were not available in Western Europe at all, and even if they had been, knowledge of Greek had all but disappeared there. The Muslim world was therefore well set by around 850 to build on the scientific traditions of Classical and Hellenistic Greeks transmitted and developed under the Roman Empire—and to acquire a centuries-long lead over Christian Western Europeans.

Complex calculations

An understanding of mathematics and astronomy is essential to the calculation of the times at which Muslims must observe their five daily prayers (times that varied widely across the vast Islamic Empire), therefore both disciplines were studied assiduously. Another, separate, intellectual tradition contributed to the development of these calculation techniques, arriving in 771 with a delegation of Hindu scholars. The scholars were visiting al-Mansur's court (which in itself illustrates the comparative openness and tolerance of the early Abbasids), and brought with them India's relatively advanced mathematics, including the use of trigonometry to help solve

The House of Wisdom played host to scholars who translated Latin and Greek works into Arabic. In doing so, they built upon classical knowledge and made breakthroughs in fields such as mathematics and medicine.

Hindu numerals, including the number zero, came from India.

Philosophical and scientific works by Aristotle and Plato came from conquered Greek lands.

Mathematical advances made possible the use of algebra and decimal places.

Arabic versions of classical Greek texts ensured the survival of ancient knowledge.

